Comptia N+ Assignment

[N+ Network Fundamental and building networks]

TCP/IP Suite

What is the TCP/IP Suite?Layers of TCP/IP: Application, Transport, Internet, and Network Access.Practical :Analyze packet headers to identify the TCP/IP layer using Wireshark.

Task: Compare OSI and TCP/IP models with real-world examples.

- **1. Open Wireshark, capture network traffic while you refresh the Flipkart homepage, and identify one packet each belonging to the Application, Transport, Internet, and Network Access layers by examining their headers.**

  - **Aim**
    - To understand the TCP/IP Suite, study its four layers, capture network traffic using Wireshark while refreshing the Flipkart homepage, identify one packet belonging to each TCP/IP layer by examining its headers, and compare the OSI and TCP/IP models with real-world examples.

  - **Introduction**
    - The **TCP/IP Suite (Transmission Control Protocol/Internet Protocol)** is a collection of communication protocols used to connect devices and transfer data over the Internet and other networks.

- It defines how data is prepared, divided into packets, addressed, transmitted, routed, and received by another device. Protocols such as HTTP, HTTPS, TCP, UDP, IP, and Ethernet work together to make network communication possible.
- For example, when we open the Flipkart website, our browser communicates with web servers through several networking protocols. Wireshark can capture and display this traffic, allowing us to examine packet headers and understand how communication takes place.

➢ **Layers of the TCP/IP Model**
➢ The TCP/IP model consists of four layers.
➢ **1. Application Layer**

- The Application Layer provides network services to applications such as web browsers, email clients, and messaging applications. It combines the functions of the Application, Presentation, and Session layers of the OSI model.
- **Examples:** HTTP, HTTPS, DNS, SMTP, and FTP.
- **Real-world example:** When we open Flipkart, the browser uses web communication protocols to request website content from the server.

➢ **2. Transport Layer**

- The Transport Layer provides communication between applications running on different devices. It can divide data into smaller units and uses port numbers to identify applications.
- Its two main protocols are:
- **TCP (Transmission Control Protocol):** Provides reliable, ordered delivery of data.

- **UDP (User Datagram Protocol):** Provides connectionless communication with lower overhead.
- **Real-world example:** TCP can carry web traffic between a browser and a web server.

➢ **3. Internet Layer**

- The Internet Layer is responsible for logical addressing and routing packets between different networks. It helps data travel from the source device to the destination device.
- **Examples:** IPv4, IPv6, and ICMP.
- **Real-world example:** IP addressing and routing help packets travel from our computer towards the servers that provide Flipkart's website.

➢ **4. Network Access Layer**

- The Network Access Layer handles communication over the local network and the physical medium. It includes functions associated with framing, MAC addresses, and transmitting data through technologies such as Ethernet and Wi-Fi.
- **Examples:** Ethernet, Wi-Fi (IEEE 802.11), and ARP in common TCP/IP model descriptions.
- **Real-world example:** When a computer uses Wi-Fi, data is transmitted through the local wireless network towards the router.

➢ **4. Tools and Requirements**

- A computer or laptop
- An Internet connection
- A web browser

- Wireshark software

➢ 5. Procedure to Capture Packets Using Wireshark

1. Open Wireshark on the computer.
2. Select the active network interface, such as Wi-Fi or Ethernet.
3. Start capturing network traffic.
4. Open a web browser and visit https://www.flipkart.com.
5. Refresh the Flipkart homepage to generate network traffic.
6. Wait a few seconds so that packets can be captured.
7. Return to Wireshark and stop the capture.
8. Examine the **Protocol**, **Source**, **Destination**, and **Info** columns in the packet list.
9. Select individual packets and inspect the packet details pane to examine the protocol headers.
10. Identify one suitable packet for each of the four TCP/IP layers and record the observations.

➢ Useful Wireshark display filters:

- http — displays packets identified as HTTP.
- tls — displays TLS protocol packets, if present.
- tcp — displays TCP packets.
- udp — displays UDP packets.
- ip — displays IPv4 packets.
- ipv6 — displays IPv6 packets.
- arp — displays ARP packets, if present.
- eth — displays Ethernet frames, if present.

- **Observation: Identifying the Four TCP/IP Layers**
  - The following table shows examples of packets that may be used to identify each TCP/IP layer. Replace the sample observations with the details from your own Wireshark capture.

TCP/IP Layer	Example Protocol or Packet	Header Details to Examine	Reason for Identification
Application	TLS or HTTP, if visible	Protocol name, request details, TLS handshake information, or application data	Represents application-level web communication
Transport	TCP	Source port, destination port, sequence number, acknowledgment number, and flags	TCP provides transport between applications
Internet	IPv4 or IPv6	Source IP address, destination IP address, and other IP header fields	IP provides addressing and routing between networks
Network Access	Ethernet II or IEEE 802.11	Source MAC address,	Represents communication

		destination MAC address, and frame type	over the local network link

- **Packet 1: Application Layer**
  - **Example protocol:** TLS or HTTP, depending on the captured traffic.
  - When a browser communicates securely with Flipkart, TLS may be visible in the packet list. If the capture contains an identifiable HTTP packet, it may also be used.
  - **Header/details to check:** Select the packet and inspect the relevant protocol details in the packet details pane. For TLS, look for handshake information if available; for HTTP, look for request methods or host information if visible.
  - **Identification:** This packet represents the Application Layer because it is associated with the web application's communication. TLS supports secure communication, although its exact mapping to TCP/IP and OSI layers is not one-to-one.

- **Packet 2: Transport Layer**
  - **Example protocol:** TCP.
  - Select a TCP packet and expand the **Transmission Control Protocol** section in the packet details pane.
  - **Header fields to check:**
  - Source Port
  - Destination Port

- Sequence Number
- Acknowledgment Number
- TCP Flags
- **Identification:** This packet belongs to the Transport Layer because TCP manages transport between applications and supports reliable, ordered data delivery.

➢ **Packet 3: Internet Layer**

- **Example protocol:** IPv4 or IPv6.
- Select a packet that contains an IPv4 or IPv6 header and expand its IP section.
- **Header fields to check:**
- Source IP Address
- Destination IP Address
- Time to Live (TTL), for IPv4
- Protocol or Next Header field, as applicable
- **Identification:** This packet represents the Internet Layer because IP provides logical addressing and routing so that packets can travel between networks.

➢ **Packet 4: Network Access Layer**

- **Example protocol:** Ethernet II or IEEE 802.11.
- Select a captured frame and expand the relevant **Ethernet II** or **IEEE 802.11** section. The available link-layer details depend on the selected interface and capture.
- **Header fields to check:**

- Source MAC Address
- Destination MAC Address
- Type or frame-control information, depending on the frame format
- **Identification:** This frame represents the Network Access Layer because it carries data over the local network link.
- **Important observation:** A single captured frame can contain multiple protocol headers. For example, an Ethernet frame may carry an IP packet, which may contain a TCP segment and application data. Therefore, the same captured packet can show information from more than one TCP/IP layer.

➢ **Comparison Between OSI and TCP/IP Models**

➢ The OSI model has seven layers, while the TCP/IP model commonly has four layers.

OSI Model	**TCP/IP Model**	**Function and Real-World Example**
Application	Application	Web browsing using HTTP or HTTPS
Presentation	Application	Data representation and encryption-related functions, such as TLS
Session	Application	Managing communication sessions between applications

Transport	Transport	TCP or UDP communication between applications
Network	Internet	IP addressing and routing between networks
Data Link	Network Access	Ethernet or Wi-Fi frames and MAC addressing
Physical	Network Access	Sending bits through cables or wireless signals

- **Main Differences**
  1. The OSI model contains seven layers, whereas the TCP/IP model commonly contains four layers.
  2. The OSI model is a general reference model for understanding network communication.
  3. TCP/IP is a practical protocol suite used for communication over the Internet.
  4. In the TCP/IP model, the Application Layer combines the roles of the OSI Application, Presentation, and Session layers.
  5. The TCP/IP Network Access Layer broadly combines the roles of the OSI Data Link and Physical layers.

- **Real-World Example: Opening the Flipkart Homepage**
- When we refresh the Flipkart homepage, the following simplified process takes place:
  1. **Application Layer:** The browser requests the website's content.

2. **Transport Layer:** TCP or UDP/QUIC carries the communication, depending on the connection and protocol used.
3. **Internet Layer:** IP addressing and routing help packets travel between networks.
4. **Network Access Layer:** The data is carried over the computer's local Ethernet or Wi-Fi connection.

- The response travels back through the network, and the browser processes the received data to display the webpage.

- **Result**
- The TCP/IP Suite and its four layers were studied. Network traffic was captured using Wireshark while refreshing the Flipkart homepage. Packet headers were examined to identify Application, Transport, Internet, and Network Access layer protocols, and the TCP/IP model was compared with the OSI model.

❖ 2. Create a table comparing the OSI and TCP/IP models, listing all layers side-by-side, and for each layer, give a real-world example from any app you use (e.g., WhatsApp message, Zomato order).

➢ Aim

- To compare the OSI and TCP/IP models by listing their layers side-by-side and explaining each layer with real-world examples from everyday applications.

➢ Introduction

- The **OSI (Open Systems Interconnection) Model** and the **TCP/IP (Transmission Control Protocol/Internet Protocol) Model** are used to understand how data is transmitted from one device to another over a network.
- The OSI model consists of seven layers, whereas the TCP/IP model commonly consists of four layers. Each layer performs a specific function in network communication.
- For example, when we send a WhatsApp message or place an order on Zomato, different networking functions work together to deliver the data to the required destination.

➢ Comparison Table of OSI and TCP/IP Models

OSI Model Layer	TCP/IP Model Layer	Function	Real-World Example
7. Application	Application	Provides network services to applications.	Sending a WhatsApp message or placing a

			Zomato order using HTTPS.
6. Presentation	Application	Handles data representation, encoding, and encryption-related functions.	A photo sent on WhatsApp is encoded into a suitable format; secure communication uses encryption.
5. Session	Application	Establishes, manages, and terminates communication sessions.	Maintaining a communication session during a WhatsApp call.
4. Transport	Transport	Transfers data between applications and uses port numbers.	TCP or UDP carries data for web communication or a voice call.
3. Network	Internet	Provides logical addressing and routes packets between networks.	IP routing carries data between your phone and the servers used by Zomato.
2. Data Link	Network Access	Transfers frames over a local network and uses MAC addresses.	Your phone sends data through a Wi-Fi connection to the router.
1. Physical	Network Access	Transmits bits using physical or wireless signals.	Wi-Fi radio signals or Ethernet cables carry network data.

- **Explanation of the OSI Layers with Examples**
- **1. Application Layer**
  - This layer provides network services used by applications.

- **Example:** When we place an order on Zomato, the application communicates with its servers to request menu details and submit the order.

➢ **2. Presentation Layer**

- This layer deals with data formatting, encoding, and encryption-related functions.
- **Example:** An image shared through WhatsApp is represented in a digital format. Encryption helps protect the contents of messages.

➢ **3. Session Layer**

- This layer manages communication sessions between applications.
- **Example:** During a WhatsApp call, communication must be maintained between the participants. In real networks, session-related functions may be distributed across several protocols and layers.

➢ **4. Transport Layer**

- This layer provides communication between applications on different devices. TCP provides reliable, ordered delivery, while UDP supports connectionless communication.
- **Example:** TCP can carry web traffic when we browse an online food-ordering application.

➢ **5. Network Layer**

- This layer provides logical addressing and routing between networks.
- **Example:** IP routing helps data travel from a mobile phone to the servers used by Zomato.

➢ **6. Data Link Layer**

- This layer transfers frames across a local network and uses hardware addresses such as MAC addresses.

- **Example:** A mobile phone sends network frames over Wi-Fi to a nearby router.

➢ **7. Physical Layer**

- This layer transmits bits through a physical medium or wireless signals.
- **Example:** Wi-Fi radio signals carry data between a phone and a wireless access point.

➢ **Real-World Example: Placing an Order on Zomato**

➢ When a user places a food order on Zomato, the networking process can be understood as follows:

1. **Application Layer:** The user selects food and submits the order.
2. **Presentation Layer:** Data representation and encryption-related functions help prepare and protect information.
3. **Session Layer:** Communication sessions are managed as the application communicates with its services.
4. **Transport Layer:** TCP or another suitable transport protocol carries the data.
5. **Network Layer:** IP addressing and routing help packets reach the destination servers.
6. **Data Link Layer:** Frames travel over the local Wi-Fi or mobile network link.
7. **Physical Layer:** Radio signals or other transmission media carry the bits.

➢ The server processes the request and sends a response back to the application, which displays the order status to the user.

➢ **Main Differences Between OSI and TCP/IP Models**

1. The OSI model has seven layers, while the TCP/IP model commonly has four layers.
2. The OSI model is a reference model used to understand network communication.
3. TCP/IP is a practical protocol suite used for communication over the Internet.
4. The OSI model separates Application, Presentation, and Session functions into three layers, while TCP/IP combines them into one Application Layer.
5. The OSI model separates Data Link and Physical functions, while TCP/IP combines them into the Network Access Layer.

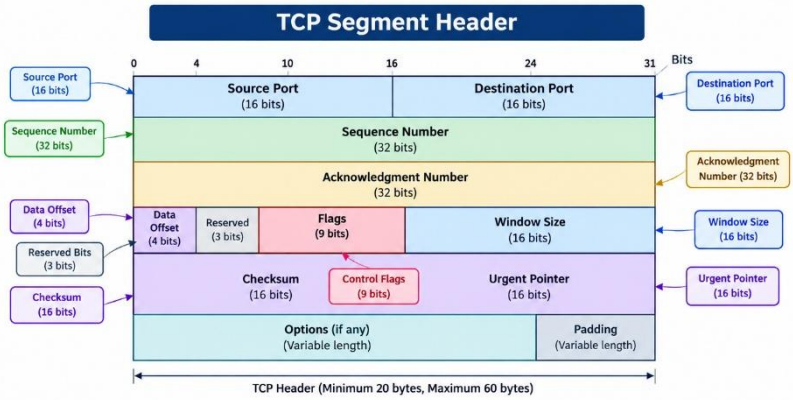
❖ **3. Given a screenshot of a TCP segment header (provided by your trainer), label the fields that belong to the Transport layer and explain what each field does in 1-2 lines. Look for fields like Source Port, Destination Port, Sequence Number, etc.**

➢ **Aim**

- To identify the fields of a TCP segment header and explain the function of each field in network communication.

➢ **Introduction**

- **TCP (Transmission Control Protocol)** is a Transport Layer protocol used for reliable communication between devices over a network. It ensures that data is delivered in the correct order and provides mechanisms to detect errors and retransmit lost data.



Source Port (16 bits)
Identifies the sending application or device.

Sequence Number (32 bits)
Indicates the position of data in the byte stream.

Acknowledgment Number (32 bits)
Shows the next expected byte from the other side (when ACK is set).

Data Offset (4 bits)
Specifies the size of the TCP header.

Reserved Bits (3 bits)
Used for future extensions (usually set to 0).

Control Flags (9 bits)
Control connection setup, acknowledgment and termination (SYN, ACK, FIN, RST, PSH, URG, etc.).

Window Size (16 bits)
Tells the sender how much data the receiver can accept (flow control).

Checksum (16 bits)
Helps detect errors in the TCP segment.

Urgent Pointer (16 bits)
Indicates the end of urgent data (when URG flag is set).

Options (Variable length)
Carries optional parameters like MSS, Window Scale, Timestamps, etc.

Padding (Variable length)
Adds extra bits to make the header a multiple of 32 bits.

Figure 1: TCP Segment Header and Its Fields

- A TCP segment consists of a **TCP header** and a **data section**. The header contains important information such as port numbers, sequence numbers, acknowledgment numbers, flags, and window size. These fields help TCP manage communication between applications.

➢ TCP Segment Header Fields and Their Functions

No.	TCP Header Field	Size	Function
1	Source Port	16 bits	Identifies the port number of the sending application. It helps the receiving device identify the application associated with the communication.
2	Destination Port	16 bits	Identifies the port number of the destination application. For example, web traffic may use destination port 443 for HTTPS over TCP.
3	Sequence Number	32 bits	Indicates the sequence position of data in the TCP byte stream. It helps TCP arrange received data in the correct order.
4	Acknowledgment Number	32 bits	Indicates the next sequence number expected from the other device when the ACK flag is set. It confirms the receipt of data up to the preceding byte.
5	Data Offset (Header Length)	4 bits	Specifies the length of the TCP header, allowing the receiver to determine where the data begins.
6	Reserved Bits	Varies by TCP specification	Reserved for future use or specified extensions. Normally, these bits should be set as defined by the applicable TCP standard.
7	Control Flags	Commonly shown as flags such as SYN, ACK, FIN, RST,	Control how the connection and data transfer operate. For example, SYN helps establish a connection, ACK acknowledges data, and FIN helps close a connection.

		PSH, and URG	
8	Window Size	16 bits	Indicates how much data the sender of this TCP segment is currently willing to receive. It supports flow control between devices.
9	Checksum	16 bits	Helps detect corruption in the TCP header and data during transmission.
10	Urgent Pointer	16 bits	Indicates the end position of urgent data relative to the sequence number when the URG flag is set.
11	Options	Variable length	Carries optional TCP parameters, such as Maximum Segment Size (MSS), Window Scale, and Timestamps.
12	Padding	Variable length	Adds extra bits, if needed, so that the TCP header ends on a 32-bit boundary.

- **Explanation of Important TCP Header Fields**
- **1. Source Port**
- The Source Port identifies the sending application or communication endpoint. It allows the destination device to associate the received data with the correct connection.
- **2. Destination Port**
- The Destination Port identifies the receiving application or service. For example, HTTPS commonly uses TCP destination port 443.
- **3. Sequence Number**
- The Sequence Number helps TCP keep data in the correct order. If data is lost, TCP can use sequence numbers to identify which data needs to be retransmitted.

- **4. Acknowledgment Number**
- The Acknowledgment Number tells the sender which byte is expected next when the ACK flag is set. This helps TCP track successful data delivery.
- **5. Data Offset**
- The Data Offset specifies the size of the TCP header. The minimum TCP header size is 20 bytes, and the maximum is 60 bytes.
- **6. Control Flags**
- Control Flags manage connection setup, acknowledgment, and termination. The SYN flag is used during connection establishment, ACK acknowledges data, and FIN indicates that a device is closing its sending direction.
- **7. Window Size**
- The Window Size supports flow control by telling the sender how much additional data the receiver is prepared to accept. This helps prevent the receiver from being overwhelmed.
- **8. Checksum**
- The Checksum is used to detect errors in the TCP segment. The receiving device verifies it to check whether the segment may have been corrupted during transmission.

- **How to Identify TCP Header Fields in Wireshark**
  1. Open Wireshark and capture network traffic.
  2. Enter the display filter tcp to display TCP packets.
  3. Select any TCP packet from the packet list.
  4. In the packet details pane, expand **Transmission Control Protocol**.
  5. Observe fields such as Source Port, Destination Port, Sequence Number, Acknowledgment Number, Flags, Window Size, and Checksum.

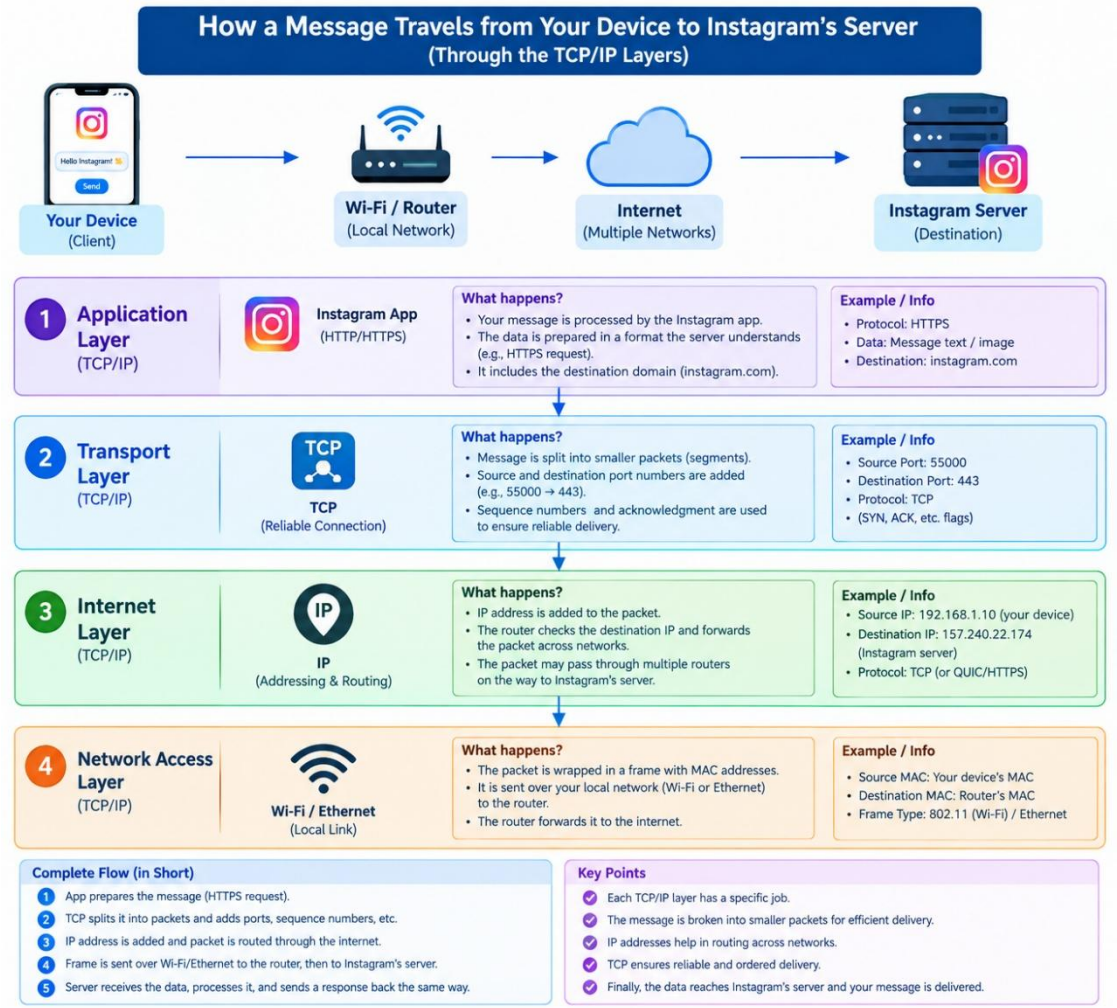
6. Match the fields with the table above and label them on the TCP header screenshot provided by your trainer.

- **Result**
- The major fields of a TCP segment header were identified, and their functions in reliable data transmission, flow control, error detection, and connection management were studied.

❖ **4. Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each step using simple app-related terms (e.g., 'message split into packets', 'IP address added', etc.).**

➢ **Message Transmission Through TCP/IP Layers**

➢ When a user sends a message using the Instagram app, the data passes through four layers of the TCP/IP model.



1. **Application Layer:** The Instagram app prepares the message and sends it using application protocols such as HTTPS.
2. **Transport Layer:** TCP divides the data into smaller segments and adds port numbers and sequence information to support reliable delivery.
3. **Internet Layer:** IP addresses are added to the packets. Routers forward the packets toward Instagram's server.
4. **Network Access Layer:** The data is placed into frames and transmitted through Wi-Fi or Ethernet to the next network device.

- At Instagram's server, the data is processed through the layers in reverse order, and the original message is delivered to the Instagram application.
- **Result:** The TCP/IP model explains how data travels from a user's device to Instagram's server through four layers.

❖ **5. Pick any one packet you captured in Wireshark and write which TCP/IP layer it belongs to, how you identified it, and what real-world action in an app (like sending a WhatsApp message or booking a ticket on BookMyShow) it represents.**

➢ **Aim**

- To identify a packet captured in Wireshark, determine its TCP/IP layer, and explain the real-world action it represents.

➢ **Introduction**

- Wireshark is a network protocol analyzer used to capture and examine network traffic. It helps us identify protocols, inspect packet headers, and understand how data travels across a network.

➢ **Packet Identification**

- For this example, a **TCP packet** is selected from the captured network traffic in Wireshark.

Particular	**Details**
Packet Selected	TCP Packet
TCP/IP Layer	Transport Layer
Protocol	TCP (Transmission Control Protocol)
Identification Method	The packet was selected in Wireshark, and the Protocol column showed TCP.

	Expanding the packet details displayed the Transmission Control Protocol header.
Important Fields	Source Port, Destination Port, Sequence Number, Acknowledgment Number, and Flags

- **How Was the Packet Identified?**
  1. Open Wireshark and start capturing network traffic.
  2. Perform an online activity, such as sending a message or opening a website.
  3. Stop the capture and enter the display filter tcp.
  4. Select a packet whose Protocol column shows TCP.
  5. Expand **Transmission Control Protocol** in the packet details pane.
  6. Observe the TCP header fields, such as source port, destination port, sequence number, and flags.
- These details confirm that the selected packet contains a TCP segment.

- **TCP/IP Layer and Real-World Application**
  - The selected TCP packet belongs to the **Transport Layer** of the TCP/IP model. TCP helps applications communicate reliably by arranging data in the correct order, acknowledging received data, and retransmitting missing data when necessary.
  - For example, when a user sends a message through WhatsApp, the message data travels over a network connection. If TCP is used for that

connection, TCP segments help transport the data reliably between the communicating endpoints. The exact protocol used can vary depending on the application and connection.

- **Result**
  - A TCP packet was identified in Wireshark by examining its Protocol column and TCP header details. It belongs to the Transport Layer of the TCP/IP model and demonstrates how network data is transported between applications.